

Topic: Embedded SQL — Self-Learning

Q. What is Embedded SQL? Discuss how SQL statements interact with host programming languages using Host Variables and Cursors. Differentiate between Static and Dynamic SQL.

📌 Definition to Remember

Embedded SQL is a technique where SQL statements are **inserted directly into the source code** of a general-purpose host programming language (C, C++, Java, COBOL). It bridges the gap between application logic and database interaction. A **pre-compiler** processes the embedded SQL before the host language compiler.

1. Why Embedded SQL?

SQL alone lacks general-purpose constructs (UI, file I/O, loops, networking). Embedding SQL in a host language allows full-featured applications to interact with databases.

2. Key Mechanisms

A. Host Variables

Variables declared in the host language (e.g., C), prefixed with a **colon (:)** when used inside SQL statements.

* **Purpose:** Bridge data between the application and the SQL query (input AND output).

```
int emp_id = 101;
EXEC SQL DELETE FROM Employee WHERE ID = :emp_id;
```

B. Cursors

A standard SQL `SELECT` returns many rows at once, but a host variable holds only one value. A **Cursor** solves this mismatch.

* Acts as a **row-by-row pointer** over a multi-row result set.

* Steps:

- | | |
|-------------------|--|
| ① DECLARE cursor | → Define the cursor + SELECT query |
| ② OPEN cursor | → Execute the query |
| ③ FETCH (in loop) | → Retrieve one row at a time into host variables |
| ④ CLOSE cursor | → Release resources |

3. Static SQL vs Dynamic SQL

Feature	Static SQL	Dynamic SQL
SQL text known at	Compile time (hard-coded)	Runtime (built as strings)
Pre-compilation	Yes — validated and optimized before execution	No — compiled at execution time
Performance	Faster (pre-optimized plan)	Slower (compiled on-the-fly)
Security	Very Secure (immune to SQL Injection)	Risky (vulnerable to SQL Injection if input is unsanitized)
Flexibility	Low — cannot adapt SQL at runtime	High — can build any query dynamically
Use Case	Standard reports, fixed operations	User-driven search screens, flexible queries

★ Must-Write Points (for 10 marks)

1. Embedded SQL = SQL statements inserted directly into a host language (C, Java, COBOL).
2. A **pre-compiler** processes embedded SQL before the host language compiler runs.
3. **Host Variables** (prefixed with `:`) bridge data between the app and SQL.
4. **Cursors** handle multi-row result sets row by row: DECLARE → OPEN → FETCH → CLOSE.
5. **Static SQL**: hard-coded at compile time; pre-optimized, secure, fast, inflexible.
6. **Dynamic SQL**: built at runtime; flexible, but slower and vulnerable to SQL Injection.
7. Static SQL is preferred for security; Dynamic SQL is used when query structure must change.

⚡ Quick Recall

Embedded SQL → Host Language + SQL → Host Variables (`:` prefix, bridges data) → Cursors (row-by-row: DECLARE→OPEN→FETCH→CLOSE) → Static (compile-time, fast, secure) → Dynamic (runtime, flexible, injection risk)